FoodAllCam – Staff Identification

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Problem

Task 1: which frames contain the staff member (person wearing the name tag)?
Task 2: staff x,y when present.

Key findings

- The tag (white rectangle with black "piano-key" pattern) is only a few pixels wide and the camera looks down, so it is rarely visible. Direct tag detection (e.g. template matching, OCR, a detector) is not reliable on this clip.
- The staff member **changes clothing appearance during the clip**: light shirt at the start, dark coat later. Detection can't be based on colour only.
- Most people are seated; only a few walk. The staff member is the sustained walker. (see Image 1)

Solution Approach

1. **Moving-object mask, any colour** – 3-frame differencing (frame t−4 frames and t+4 frames (about 0.16 s), both must change) AND difference from a median background.
    a. The model will respond to *walking people* only, not *seated ones*.
    b. This will be therefore also independent of clothing colour.
2. **Moving Humans → Short path** – a person-sized moving item linked frame-to-frame with a strict jump limit
    a. This avoids the tracking to hop between different people.
3. **Starting point** – the short path with the longest walked path to find the sustained walker.
4. **Chain across the clothing change** – link the short paths backwards/forwards from the starting point, i.e. when the next one starts close in space (≤90 px) and time (≤8 s).
    a. Identity is carried by *position/time continuity*, not colour.
    b. Therefore, we can detect the same person even when he changes shirt from light-shirt to dark-coat.
5. **Gap filling** – when the model loses the person for a short time, it guesses where they were during that time instead of leaving a hole. (see Image 3)
    a. Done by drawing a straight line between those two points and placing the person at evenly spaced positions along it.
6. **Output** – Results are saved as a per-frame table (presence + position), a marked-up video, a timeline, and a path image.

Result on sample.mp4

- Staff present: **roughly 12.6 s – 51.1 s (frames 314–1277)**
- The light-shirt phase (around 13–26 s) is tracked well; the box follows the light-shirt walker up and down the corridor. (see Image 2)
- Around 26–32 s, staff stops/slows at the bottom of the corridor and puts on the dark coat; position is estimated (interpolated). (see Image 3)
- Around 32–51 s, the dark-coat phase followed by his movements. (see Image 4)

Assumptions and limitations

- One staff member; identity is assumed continuous because his paths start where the previous one ended.
- Accuracy is judged visually. In the dark-coat phase (around 34–48 s) several dark-clothed people move near the staff, so the chain may attach to the wrong person at times.
- Interpolated stretches are estimates (slow/stationary person produces no motion for the differencing step).
- Global median background: slow lighting change or long-stationary staff would be lost.

Appendix



Image 1: Employee with Tag is spotted at 13s.

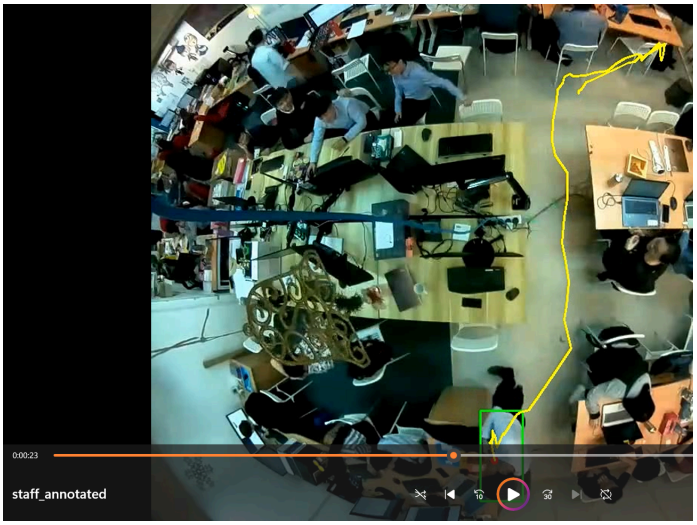


Image 2: Employee's path is followed

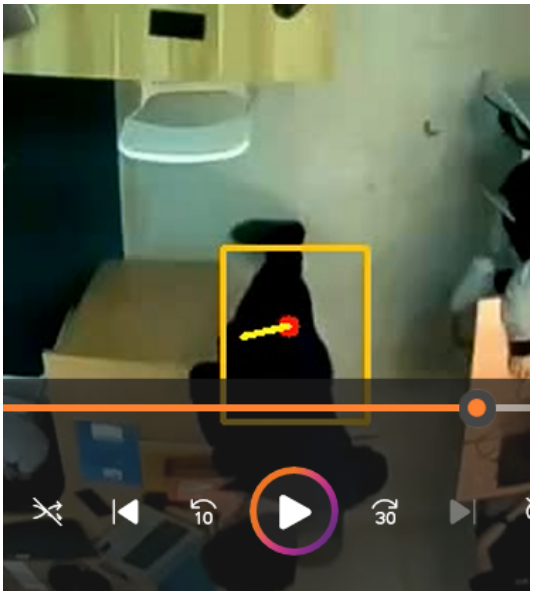


Image 3: Employee's path is stationary

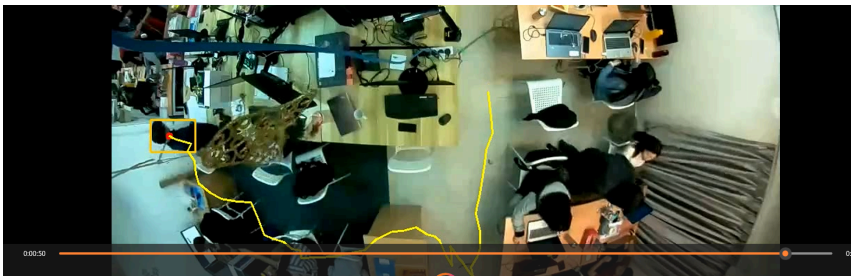


Image 4: Employee is tracked.